Ex – 03	EDA – Data Import and Export

AIM

To import data from CSV files, Excel files, SQL databases, and web sources using Python, handle different data formats, and export a Pandas DataFrame to an Excel file.

PROBLEM STATEMENT

To demonstrate data import and export techniques in Exploratory Data Analysis (EDA) using Python. The program reads data from different sources and formats, processes the data using Pandas, and exports the resulting DataFrame into an Excel file for further analysis.

PROCEDURE

1. Import the required libraries, including Pandas, SQLite3, and Requests.
2. Create a sample dataset using Pandas DataFrame.
3. Export the dataset to a CSV file and import it using read_csv().
4. Export the dataset to an Excel file and import it using read_excel().
5. Create an SQLite database and store the dataset in a database table.
6. Import the data from the SQL database using read_sql_query().
7. Retrieve sample data from a web source using the Requests library.
8. Handle different data formats and display the imported datasets.
9. Export the final DataFrame to an Excel file using to_excel().
10. Verify the exported file and display the successful completion message.

```
# EDA - Data Import and Export

import pandas as pd
import sqlite3

# 1. Create a sample DataFrame
data = {
    'ID': [1, 2, 3, 4, 5],
    'Name': ['Arun', 'Bala', 'Cathy', 'Divya', 'Ezhil'],
    'Department': ['IT', 'CSE', 'IT', 'ECE', 'CSE'],
    'Marks': [85, 90, 78, 88, 92]
}

df = pd.DataFrame(data)

# 2. Export data to CSV and Excel
df.to_csv('students.csv', index=False)
df.to_excel('students.xlsx', index=False)

# 3. Import data from CSV
csv_data = pd.read_csv('students.csv')
print("CSV Data:")
print(csv_data)

# 4. Import data from Excel
excel_data = pd.read_excel('students.xlsx')
print("\nExcel Data:")
print(excel_data)

# 5. Import data from SQL database
conn = sqlite3.connect(':memory:')
df.to_sql('students', conn, index=False, if_exists='replace')

sql_data = pd.read_sql_query(
    "SELECT * FROM students", conn
)
print("\nSQL Data:")
print(sql_data)

# 6. Import data from a web source
url = 'https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv'
web_data = pd.read_csv(url)
print("\nWeb Data (First 5 Rows):")
print(web_data.head())

# 7. Handle different data formats
print("\nData Types:")
print(df.dtypes)

# 8. Export DataFrame to Excel
df.to_excel('output_students.xlsx', index=False)
print("\nData exported successfully to output_students.xlsx")

conn.close()
```

```
CSV Data:
   ID   Name Department  Marks
0   1   Arun         IT     85
1   2   Bala        CSE     90
2   3  Cathy         IT     78
3   4  Divya        ECE     88
4   5  Ezhil        CSE     92

Excel Data:
   ID   Name Department  Marks
0   1   Arun         IT     85
1   2   Bala        CSE     90
2   3  Cathy         IT     78
3   4  Divya        ECE     88
4   5  Ezhil        CSE     92

SQL Data:
   ID   Name Department  Marks
0   1   Arun         IT     85
1   2   Bala        CSE     90
2   3  Cathy         IT     78
3   4  Divya        ECE     88
4   5  Ezhil        CSE     92

Web Data (First 5 Rows):
   sepal_length  sepal_width  petal_length  petal_width species
```

```
0          5.1          3.5           1.4          0.2  setosa
1          4.9          3.0           1.4          0.2  setosa
2          4.7          3.2           1.3          0.2  setosa
3          4.6          3.1           1.5          0.2  setosa
4          5.0          3.6           1.4          0.2  setosa

Data Types:
ID             int64
Name          object
Department    object
Marks          int64
dtype: object

Data exported successfully to output_students.xlsx
```

RESULT

Data was successfully imported from CSV, Excel, an SQLite database, and a web source using Python Pandas. The data was processed and exported to final_students.xlsx, demonstrating the basic data import and export operations used in Exploratory Data Analysis (EDA).